

Task-Specialized Harness Refinement Across Financial Decision Tasks

Anonymous ACL submission

Abstract

Financial document agents need more than valid structured outputs: preprocessing can omit relevant text, and an evidence-linked judgment can still misinterpret a clause. We investigate task-specialized harness refinement through Fin-Harness, a preprocessing, analysis and evidence-validation pipeline for Korean insurance, US credit cards and US commercial loans. We fix a roster of 1,000 sources per task and measure offline preprocessing on all 3,000, including extraction-review and parser-unsupported inputs. Original rule filters preserve most source-present taxonomy keywords, whereas the original Korean TXT parser returns no clauses for 210 sources. A three-role preprocessing loop selects configurations on 704 development-linked inputs before applying them to the full cohort; additional size reductions are predominantly whitespace. Twelve original-task development documents expose a second bottleneck: source-conditioned model grades confuse absent categories with missed evidence and can contradict their own taxonomy judgments. Prespecified controls therefore block automatic detector refinement. A separate loan-reference repair raises visible paragraph/category evidence retention from 80.7% to 100%, without establishing relevance. Reusing identical-pool preparation reduces local first-pass retrieval time by 54.3% on 1,508 diagnostic queries without changing returned rows. These pilot components motivate distinct execution and semantic gates; full-cohort downstream comparisons and human validation remain incomplete.

Working manuscript. Verified component results are separated from non-evidentiary historical tables preserved in the appendix.

1 Introduction

Financial agreements pose a practical engineering problem for language-model agents: an ap-

parently complete report can be based on incomplete input. An insurance product summary, a multi-column card agreement and a commercial-loan amendment differ in layout, language, scope and the evidence needed to review a finding. A JSON schema checks a report’s shape, not whether the parser retained an exception or the analyst identified the right risk.

We study the *harness*: the software and task instructions that determine what the model receives, which tools it uses and which artifacts may proceed to the next stage. Harness optimization has been investigated as an outer-loop search over code and execution feedback (Lee et al., 2026; Lou et al., 2026). In financial review, however, a high stage-local score is useful only if its measurement process is fit for the decision being optimized. A system can improve a lexical retention score without improving legal recall, or optimize a judge that rewards a mislabeled finding.

Fin-Harness separates document preprocessing, risk-candidate analysis and evidence validation. Its proposed outer loop combines Meta-review, Evolution and Generation roles without changing model weights. Our current empirical contribution is a reproducibility and failure-analysis study of that design, not a demonstration that three agents outperform one. We ask: (1) what reaches the model when the original preprocessing is applied to a fixed financial corpus; (2) what an executed, bounded preprocessing loop changes; and (3) whether source-conditioned grades are ready to select detector revisions.

The study contributes a failure-inclusive 3,000-source roster and offline comparison, an auditable Stage 1 refinement trajectory, and a diagnostic showing why schema acceptance and literal evidence binding must remain separate from semantic evaluation. The intended application is analyst-assisted document triage. We have not established production deployment, analyst time savings or re-

liable financial decisions.

2 Related Work

LEGAL-BERT studies domain adaptation for legal NLP (Chalkidis et al., 2020); CUAD provides expert annotations for salient contract review spans (Hendrycks et al., 2021). Our source roster is neither a replacement for expert labels nor a directly comparable contract-understanding benchmark. It includes heterogeneous documents and unresolved extraction failures.

Meta-Harness searches harness code using prior candidates’ source, scores and traces (Lee et al., 2026). AutoHarness synthesizes code with environment feedback to constrain agent behavior in games (Lou et al., 2026). We investigate a domain-specific engineering difficulty: the feedback itself can fail before optimization is meaningful. Our bounded typed configuration search is narrower than general code synthesis.

LLM-as-a-judge research identifies position, verbosity and self-enhancement biases, alongside reasoning limitations (Zheng et al., 2023). Agreement measured on chat preferences does not establish agreement with legal experts. Here, detector and judge request the same model alias, so independence and mitigation of self-preference are not claimed.

3 System and Refinement Protocol

3.1 Original tasks and executed transport

Figure 1 preserves the original design. Korean insurance analysis uses the original TXT-harness parser, risk taxonomy, validation resources and severity logic. The parser recognizes summary-style headings, drops short clauses and caps each retained clause at 1,600 characters. This is a measurable input restriction, not a guarantee of relevant-content preservation. Card and loan analysis reuse their original task prompts, taxonomies and algorithms through explicit full-source windows: at most 18,000 characters with source units and overlap. For loan trigger quotation, the adapter selects one source unit of at most 240 characters to respect the original 300-character quotation constraint.

The current execution requests gpt-5.4-mini through authenticated Codex CLI subscription transport. This is an alias, not a pinned model snapshot. We preserve original task logic while disclosing transport, structured range selection

and windowing adaptations; byte-identical execution of every original command is not claimed. The earlier generic pilot’s task and scoring algorithms are not substituted for these tasks. Only its transport/storage utilities are reused.

Every stored call binds the request, schema, native response and observed usage to its source and implementation snapshots. The host materializes selected text ranges from the actual source rather than accepting a model-retyped quotation. Evidence identifiers are checked against the relevant resources. These operations establish traceability, not entailment, legal relevance or truth. Loan pilot retrieval uses analogous contract clauses, not court precedents; card search results are not expert relevance judgments.

3.2 A bounded, stage-local search

Let b_d be the original rule-filter output for document d , and $f_\theta(b_d)$ a candidate transformation. The executed Stage 1 objective is the lexicographic minimization

$$\min_{\theta} \left(\sum_{d \in D_{\text{dev}}} W(f_\theta(b_d)), \sum_{d \in D_{\text{dev}}} |f_\theta(b_d)| \right), \quad (1)$$

subject to per-document gates, where W counts whitespace-delimited units and $|\cdot|$ counts Unicode characters. Neither quantity is billed model tokens. Gates preserve baseline inventory keywords and configured financial-pattern occurrence counts, retain at least 80% of baseline whitespace units, and require pairwise lexical cosine of at least 0.99 when defined. Loss of a previously defined vector is rejected.

At each iteration, Meta-review reads the full previous attempt history; Evolution reads that review, the current state and measured outcomes; Generation reads both outputs and proposes typed configurations. Available operations are literal line/prefix removal, blank-line/space collapse and separator-line removal, not arbitrary code generation. The deterministic evaluator records every proposal, including duplicates. A novel eligible candidate is retained only if it strictly improves Eq. 1; the loop stops on no improvement or a three-iteration per-domain cap.

This is an operational decomposition, not a proof that local objectives factorize an end-to-end financial utility function. Stage 2/3 refinement, agent-removal ablations and general code/hook evolution remain unexecuted. We make no claim

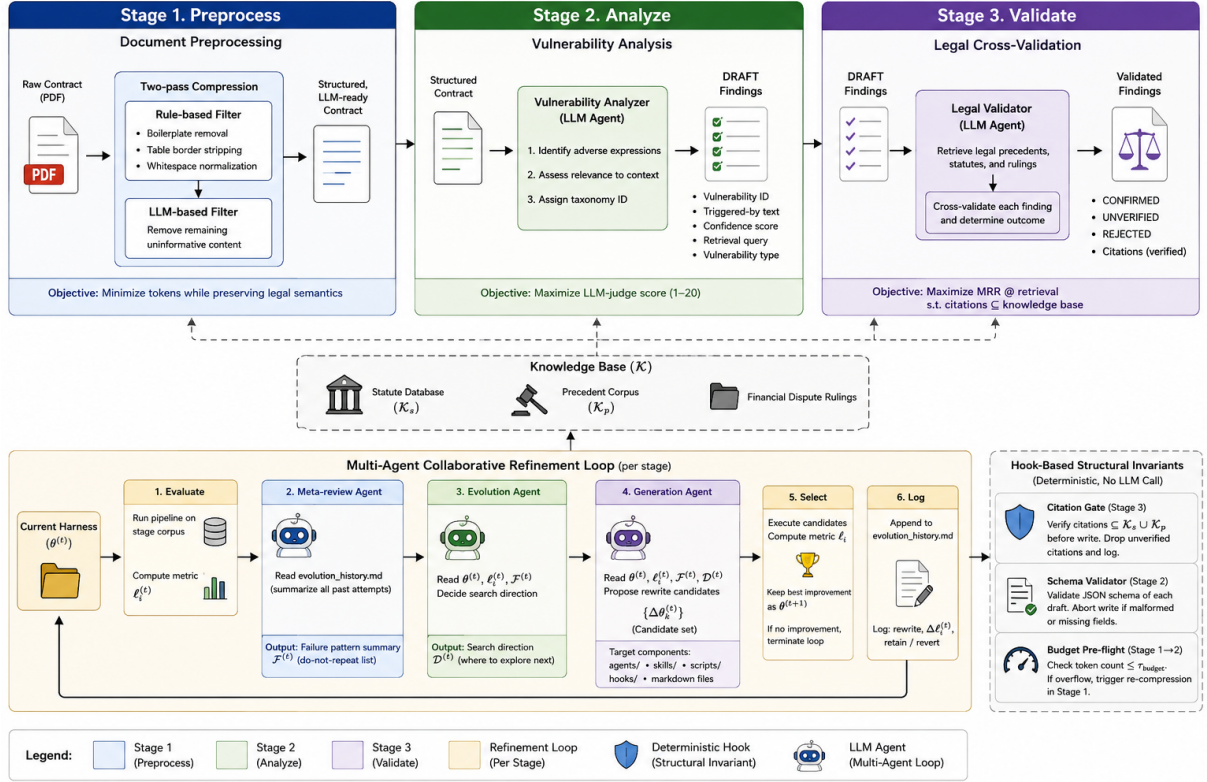


Figure 1: **Original reference design, not a completed execution trace.** The current run implements the three-role loop only for deterministic Stage 1 configurations. The pictured per-document LLM filter, Stage 2/3 optimization and token-overflow recompression are not established by these experiments. A citation’s presence in a repository does not establish legal validity.

that the three-role design eliminates duplicate proposals, escapes local optima or guarantees convergence.

4 Fixed Cohort and Experimental Scope

The main denominator is fixed at 1,000 sources per task (Table 1), not reduced to successful executions. Korean sources include product summaries and short examples; SEC loan filings include amendments and commercial agreements, not only complete consumer contracts. The card collection includes 25 distinct PDFs from an accessible current CFPB issuer page after an archive request failed. They are not relabeled as archive-vintage documents. Twelve card filenames have Spanish indicators, and two inspected texts are Spanish; English-only coverage is not verified.

PyMuPDF extraction marks 2,618 rows mechanically successful and 382 for review. These flags describe extraction, not content completeness. The original KR parser additionally yields no clauses for 38 successful extractions. Consequently, 2,580 sources are current runtime candidates; the remaining 420 stay in the roster for resolution, not as automatically assigned zero-accuracy outcomes. An OCR diagnostic produced alternate text, but unresolved reading-order fidelity prevents treating it as canonical.

Provisional source-family and text-similarity grouping marks 704 inputs development-linked (611 KR, 71 card, 22 loan), leaving 2,296 evaluation-reserved. These are exposure strata, not verified independent held-out splits. Stage 1 selection uses only the 704 development-linked inputs, including their quality flags; all domain configurations are frozen before full-cohort application. Offline E1 reports all 3,000 sources. The original-task diagnostic uses four fixed development sources per domain, not 12 replacements for the main cohort.

5 Results

5.1 E1: Full-cohort preprocessing

We compare raw extraction, unchanged original rule filters and an explicit float64 adaptation of each original rank method. KR/loan use a TF-cosine PageRank graph; card uses pair-local TF-IDF degree scores, not PageRank. Nine bounded cases match original reference outputs; full-cohort bitwise equivalence is not established. The KR TXT parser is evaluated separately from its rule

filter. BART rows remain not run (NR), without substitute heuristics or inherited scores.

For source d , output o and source-present unique inventory keywords $K(d)$, we compute $|o|/|d|$ and $|K(d) \cap K(o)|/|K(d)|$, together with pairwise TF-IDF lexical cosine. Ratios are macro-averaged over their defined document denominators, which differ by metric. Empty denominators remain null, not perfect preservation or an omitted source. These metrics do not measure semantic fidelity, legal recall or inference cost.

The original rules retain 97.42%, 91.86% and 95.06% of characters for KR/card/loan, respectively, with source-present keyword retention 99.96%, 99.60% and 99.98%. This is limited compression with high lexical preservation, not evidence of useful downstream savings. Rank methods retain fewer characters and more often lose inventory keywords. Their values do not show which removed passages were legally important.

The original KR parser yields no clauses for 210/1,000 sources. Its mean keyword retention is 71.86%, and its 40.20% document-macro character ratio differs from 35.6% pooled literal coverage. At least one returned clause reaches the 1,600-character cap in 746 sources. Thus, the parser payload and the nearly lossless original rule-filter baseline are materially different representations; they cannot share a preprocessing-fidelity claim.

5.2 Executed Stage 1 refinement

The three-role loop makes 21 calls with 284,431 observed tokens. Nine proposals include six novel evaluated configurations, three duplicates and five selections. No evaluated proposal violates a gate. KR stops at its cap; card/loan stop because their last proposed candidates do not strictly improve the objective. These stopping conditions neither prove global optimality nor support a ten-iteration convergence claim.

Table 3 separates selection from full-cohort application: KR removes only three whitespace units in development and 17 overall; card and loan remove none. Character reductions are mainly whitespace. All full-cohort gates pass against the original filter, but the prior extraction flags remain. Passing a relative gate cannot repair defective input, establish legal preservation or improve the separate KR parser.

Task	Source collection / document scope	Selected	Extracted	Review
T1 (KR-Ins)	Korean insurer documents, including product summaries	1,000	761	239
T2 (US-CC)	CFPB card agreements, including current-page supplement	1,000	858	142
T3 (US-Ln)	SEC EDGAR commercial-loan filings, including amendments	1,000	999	1
Total	Fixed source roster; no success-only reduction	3,000	2,618	382

Table 1: **Verified fixed-cohort counts for the three financial tasks.** “Extracted” denotes mechanical text-extraction success, not source completeness, independence, legal validity or completed model analysis. Review rows remain in the selected denominator. The original 600/800/223 counts are not used for E1.

Dom.	Strategy	Char.	KW ret.	Lex. cos.
KR-Ins	Raw	1.0000	1.0000	1.0000
	Rank*	0.6012	0.6083	0.8901
	BART	NR	NR	NR
	Orig. rule	0.9742	0.9996	0.9971
	TXT parser	0.4020	0.7186	0.8271
US-CC	Raw	1.0000	1.0000	1.0000
	Rank*	0.7268	0.8847	0.9776
	BART	NR	NR	NR
	Orig. rule	0.9186	0.9960	0.9912
US-Ln	Raw	1.0000	1.0000	1.0000
	Rank*	0.8411	0.9286	0.9925
	BART	NR	NR	NR
	Orig. rule	0.9506	0.9998	0.9957

Table 2: **E1 — Verified offline measurements on all 3,000 sources.** Each domain retains 1,000 inputs, including extraction-review rows. Char./keyword denominators: KR 998/887, card 980/921, loan 1,000/929. Lexical denominators: KR 998 (parser 790); card 939 (rule 938); loan 1,000. *Explicit rank adaptation. NR: not run. No accuracy or inference-cost claim.

5.3 Detector and grader diagnostics

The 12-document original-task pilot contains 96 draft findings: 13 KR, 82 card and one loan. All 96 quotations match their actual source; two KR drafts are rejected by the original validator, leaving 94 final report rows. These are traceability checks, not legal accuracy or first-attempt success. Three loan outputs are empty and remain in the diagnostic. The full-source grader receives all 227,568 extracted characters, not only detector-selected passages, and scores trigger fidelity, description, query quality and coverage on adapted 1–5 axes. For empty outputs the first three axes are null, not invented zero scores.

The initial grader accepts all 96 findings as supported and proposes 75 possible omissions. These are model labels, not precision or recall. Inspection finds that all five loan omission entries on one document describe an *absent* risk category rather than missed positive evidence. Literal source linkage has not prevented a semantic error.

Task	Development		All 1,000/task	
	ΔW	Δ chars	ΔW	Δ chars
KR-Ins	3	151,635	17	205,371
US-CC	0	0	0	0
US-Ln	0	221	0	179,199

Table 3: Additional units/characters removed relative to the original rule filter by the frozen Stage 1 selection. Positive values denote reductions, not model-token savings. Development sizes are 611/71/22; all-source application is 1,000 per task and includes development sources.

Grader	Calls	Controls passed	Real regrades
v1	18	13/18	0
v2	18	17/18	0
v3 (bound)	24	23/24	0
v4 (schema)	24	22/24	0

Table 4: Exposed, assistant-authored diagnostic controls. Every control must pass its frozen gate before 12 real regrades can run. Fractions are not independent accuracy estimates; versions are outcome-informed revisions. The rejected v3 answer remains in the denominator.

We therefore run frozen synthetic controls before allowing regrades to select a new detector. Each task tests no issue, missed issue, correct finding, unsupported claim, irrelevant query and wrong taxonomy. V3 adds three new correct/wrong-label pairs and binds each finding to its literal original taxonomy definition.

V2 misreads a card finding’s declared taxonomy. V3 echoes the correct declared ID and identifies its mismatch, yet returns both supported and `does_not_fit`. The host rejects this contradiction without repairing its label or score. V4 excludes the same inconsistent combinations in the generation schema, without changing V3’s prompts, controls or score bounds. All 24 grades are structurally accepted, but two KR controls fail. A missed-issue case receives coverage 5 despite a rationale stating that no issue was captured. For

this one case, recorded model alias, system, input, schema and output target match V3, which returned coverage 1. This paired disagreement is not a failure-rate estimate or evidence that the schema caused a regression; the alias does not pin hidden backend conditions.

The second case receives partial support for a waiting-period finding. Its rationale attributes ambiguity to a title that makes no such claim, while the taxonomy definition itself does mention unclear conditions. This exposes control/rubric ambiguity as well as a grading concern; independent adjudication is needed. Table 4 retains both failures without revising expectations. No version clears its selection gate, and no follow-up real regrade runs. Neither these development-exposed fractions nor a future pass would establish independent legal accuracy.

5.4 Reference-window repair

The original loan pool tags full paragraphs but stores only their first 1,500 characters. Across all 1,000 loans, only 2,351/2,913 original paragraph/category pairs retain visible keyword evidence (80.7%). Exact keyword-bearing windows restore all 2,913 pairs. A compact window subset selected on the existing 22 development loans retains these pairs with 10.7% fewer indexed characters than full windows (Appendix B). It can omit other same-category clauses; no new pool is admitted to the runtime on lexical gates alone. These are manually engineered repairs, not an executed Stage 3 outer loop.

Separately, reusing pool tokenization and BM25 construction preserves the original query expansion, query-dependent IDF and MMR. On 1,508 diagnostic queries from 883 loans, the original pool’s first-pass retrieval time falls from 35.30 to 16.14 seconds including cold preparation (54.3%). Across three pools and three repetitions, all 13,572 paired results match exactly. These are shared-desktop retrieval times, not downstream accuracy or end-to-end model latency (Appendix B).

6 Operational Implications

The evidence identifies distinct release gates. First, retain source and failure denominators so extraction success cannot conceal parser loss. Second, verify exact requests, source ranges and native responses so a score is tied to the intended task. Third, validate the semantic evaluator before op-

timizing against it. These are engineering lessons from the recorded failures, not measured benefits in deployment.

The current execution envelope cannot fund full-cohort inference (Appendix A). We therefore report pilot components separately from the retained 3,000-source roster. Completion requires budget reconciliation and a fit evaluator, not a success-only dataset.

The proposed downstream comparison must keep source identity, input coverage, model settings and quality strata matched across configurations. Agent-removal ablations need comparable call budgets and independent repeated runs. Human review must establish positive findings and missed risks with recorded annotation/adjudication, rather than relabeling model grades as gold. These remain requirements, not completed experiments.

7 Conclusion

Fin-Harness exposes a practical boundary between executable harnesses and trustworthy optimization. Full-cohort preprocessing reveals parser loss that a rule-filter score would hide; executed Stage 1 refinement mainly changes whitespace; source-linked grades fail semantic controls; source-window repair fixes a lexical reference defect; preparation reuse reduces local retrieval time while preserving identical-pool outputs. We retain the full 3,000-source scope while withholding downstream comparative, convergence and human-precision claims that the current records do not establish.

Limitations

The study does not complete the requested full-cohort model analysis, Stage 2/3 refinement, baseline comparisons, repeated-seed ablations or expert validation. There is no demonstrated deployed-user or financial outcome benefit. The original broad claim that a single refinement loop outperforms all baselines is not supported by the current executions.

Source diversity limits interpretation: summaries, examples, amendments, language mixtures and extraction-review cases are not interchangeable complete contracts. Provisional group and duplicate screening does not establish issuer independence or a held-out population. Full-cohort Stage 1 application includes development-linked sources and is reported as such. Extracted character and lexical-retention measures do not quantify preserved legal meaning or downstream error. The reference-window repair retains category keywords, not every clause occurrence or exception. Its motivating defect was observed in earlier all-source diagnostics; reserved inputs are not untouched held-out data.

The bounded mutation space restricts what Stage 1 can discover. Three-role execution is not an ablation proving that all roles are necessary. Synthetic controls are simple, author-exposed and revised after observing failures. The detector and judge share a requested model alias; schema and consistency guards cannot eliminate same-model bias or establish calibration on natural documents. No statistical uncertainty interval is offered as if these development cases were an independent random sample.

Original-reference rank agreement is bounded, and floating-point rounding/ties may affect ordering. The transport preserves native records but does not pin the provider model snapshot. Observed-token accounting has two older records with unknown usage and no exact dollar invoice. Historical numerical tables in Appendix C are preserved solely for author reconciliation and cannot support any finding here.

Ethical Considerations

The intended use is review assistance, not automated legal advice, insurance eligibility, lending or adverse financial action. An adverse clause is not automatically unlawful; relevance depends on contract context and applicable authority. Source-

linked quotations and authority IDs do not establish a defensible legal conclusion. Qualified human review is needed before consequential use.

Only public-document research is described, with synthetic user profiles rather than an observed customer study. Public availability does not establish unrestricted redistribution rights. Source licensing and sensitive-content review must precede release of full documents or outputs. No consent process, human-participant study or expert adjudication is claimed. The review manuscript contains no identity-revealing repository or private-storage link.

References

- Ilias Chalkidis, Manos Fergadiotis, Prodromos Malakasiotis, Nikolaos Aletras, and Ion Androutsopoulos. 2020. [LEGAL-BERT: The muppets straight out of law school](#). In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 2898–2904, Online. Association for Computational Linguistics.
- Dan Hendrycks, Collin Burns, Anya Chen, and Spencer Ball. 2021. [Cuad: An expert-annotated nlp dataset for legal contract review](#). *Preprint*, arXiv:2103.06268.
- Yoonho Lee, Roshen Nair, Qizheng Zhang, Kangwook Lee, Omar Khattab, and Chelsea Finn. 2026. [Meta-Harness: End-to-end optimization of model harnesses](#). *Preprint*, arXiv:2603.28052.
- Xinghua Lou, Miguel Lázaro-Gredilla, Antoine Dedieu, Carter Wendelken, Wolfgang Lehrach, and Kevin P. Murphy. 2026. [Autoharness: improving llm agents by automatically synthesizing a code harness](#). *Preprint*, arXiv:2603.03329.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. [Judging LLM-as-a-judge with MT-Bench and chatbot arena](#). *Preprint*, arXiv:2306.05685.

A Metric and Artifact Details

Exact E1 scope. The offline pass stores 3,000 document result records, 7,000 transformed texts and 3,000 references to raw extraction: raw/rule/rank for each task and a separate KR parser payload. BART and other unexecuted model compressors are NR, not reconstructed from prior paper values. All metrics retain source identity and quality/exposure strata.

Source-present keyword retention deduplicates the original inventory after normalization (KR case-sensitive; US lowercase), then counts literal substring presence. It differs from the legacy inventory hit rate, which keeps repeated inventory entries and includes absent keywords in its denominator. Lexical cosine uses pair-local TF-IDF over lowercase alphanumeric tokens of length at least two, additionally allowing Hangul for KR. Empty token vectors are null.

For every method, character ratios are defined on 998 KR, 980 card and 1,000 loan sources; source-present keyword retention on 887, 921 and 929. Lexical counts are 998 for KR raw/rule/rank and 790 for its parser, 939 for card raw/rank and 938 for its rule filter, and 1,000 for loan. A null result remains an explicit record. Aggregate means do not silently impute zero or one.

Original task modules. The executed insurance source is ECC_harness_v3_txt, including its original parser, task instructions, resources and severity logic. US tasks come from eval_US_card and eval_US_loan. Selected symbols are loaded without invoking the original paid-API initializers. The current evolve.py routes to the executed Stage 1 runner in scripts_evolve; the earlier stub is preserved separately. A prior generic pilot is not used as the current detector or evaluator.

Trace and budget semantics. Saved native records include exact system/payload/schema, selected source ranges, output and usage, with hashes binding source and implementation snapshots. Audits reconstruct original detector payloads and materialized findings before assigning a grade to a document. Failed native responses remain in accounting. Cached responses are reused only after identity verification, not counted as fresh model experiments. Prelaunch observed-token stops cannot enforce a hard provider token ceiling on the last call.

The Stage 1 trace contains 7,622 document-

output records: 4,622 development incumbent/candidate evaluations and 3,000 frozen final applications. Full-cohort application does not feed back into selection. The initial 12 full-source grades use 407,743 observed tokens; control v1/v2 together use 473,152, and v3 uses 325,363. At the latest diagnostic cutoff, v4 adds 326,769 tokens across 24 calls. The scoped inventory contains 619 stored calls and 16,112,787 known observed tokens, plus two older calls with unknown usage. These counts include earlier engineering diagnostics; they are neither a per-system benchmark nor account-wide billing.

A current one-configuration preflight needs at least 9,578 native calls for the 2,580 runtime candidates, plus up to 858 conditional card-validation calls. This excludes recovery, baselines, refinement and independent evaluation; it does not shrink the required 3,000-source main run. The recorded 1,100-call/30-million-observed-token development envelope cannot support that plan. Observed subscription tokens are not an API invoice, and missing historical usage is not zero.

Required completion evidence. A final experiment must resolve or explicitly disposition every fixed source, retaining its source ID and failure cause, and report quality and development-exposure strata. It must validate a semantic evaluator before automatic selection, execute the remaining comparisons under reconciled budgets, and tie each table cell to native runs and annotation evidence. The current manuscript is an interim, evidence-correct working version. Its formatting and component audits do not certify study completion.

B Pilot Source and Reference Audits

The latest four completed originals per task yield 96 draft flags and 94 final report rows. Source and extracted-text hashes for all 12 documents, 40 stage-artifact hashes and all 96 literal quotations were rechecked. The final KR rows link to retained validation records; two rejected drafts remain in the ledger. Latest completion is not pass@1: an earlier loan failure is preserved and the current loan outputs follow a recorded rerun.

Table 5 preserves original paragraph eligibility and the first-40-eligible-paragraph cap. Full windows are exact source spans of at most 1,500 characters with 24-character overlap, retaining only keyword-bearing windows. Compact cover

Metric	Original	Full windows	Compact cover
Original paragraph/category pairs	2,913	2,913	2,913
Pairs with visible category keyword	2,351	2,913	2,913
Pair retention (%)	80.707	100	100
Unsupported emitted tag assignments	562	0	0
Reference chunks	2,572	2,922	2,652
Indexed characters	2,676,497	2,984,279	2,665,494
Union of covered source characters	2,676,497	2,976,407	2,663,862

Table 5: Verified lexical reference audit across all 1,000 loans, including 129 original zero-chunk sources and one quality-flagged input. Repeated windows do not inflate the fixed paragraph/category denominator. Neither keyword retention nor indexed size is relevance or measured latency.

Pool	Fresh (s)	Prepared (s)	Reduction
Original	35.301	16.145	54.27%
Full windows	38.546	16.841	56.31%
Compact cover	35.536	16.264	54.23%

Table 6: First-pass retrieval over all 1,508 queries. Prepared time includes all cold scoped-index builds. Each row compares an identical reference pool; it does not assess relevance or model inference.

greedily selects windows covering all original parent categories; ties favor earlier source positions. Its complete-retention and visible-tag gates select the smallest development index before application to the 978 reserved loans. Development indexed characters are 79,667/93,220/79,856 for original/full/compact: compact is slightly larger than original there. Over all loans it saves only 0.41% against original, versus 10.68% against full windows, and adds 80 chunks. All 8,146 candidate chunks were checked against their exact raw source spans. These measurements justify a literal-evidence repair, not semantic completeness, new relevance labels, runtime pool adoption or an automatic improvement curve.

Prepared retrieval. The original extractor generates at most three taxonomy-triggered queries per source: 883 loans yield 1,508 queries and 117 zero-query sources remain explicit. There are 31 development-linked and 1,477 reserved queries, not independent relevance labels. Only successful references from the 22 development loans are admitted, excluding each query and its provisional source component. Original/full/compact pools contain 68/82/70 chunks. Ten distinct scoped indexes per structure are constructed once; query results are never cached. Structure order rotates and arm order alternates over three repetitions with a fixed Python hash seed.

All 13,572 original/prepared pairs preserve ex-

act returned rows and repeat stability, with zero execution errors, exclusion violations or fallback calls. Independent saved-record checks re-hash 2,000 source/text files and recompute latency aggregates. Original-pool p50/p95 fall from 23.151/26.516 to 10.570/12.112 ms. Cold totals are 159/175/158 ms. Table 6 reports retrieval-only timing on one shared Windows desktop: PDF compilation overlaps 14.24 seconds of the timed phase. All observations are retained; no isolated-host or end-to-end claim is made. Cross-structure source context changes in 1,468/1,508 original-full and 1,221/1,508 original-compact queries. Thus exact prepared-serving agreement does not justify swapping reference pools. The prepared component is opt-in; the frozen main runner remains unchanged. There are no additional model calls.

C Preserved Historical Material: Not Evidence

The original figures and tables remain as an *author-reconciliation archive*, not empirical support. All E2–E5 values and plotted coordinates are unchanged. Their source, model-run, relevance and annotation traces are not established on the fixed 3,000-source cohort.

In particular, the old KR iteration endpoints (18.9/20 and MRR 0.792) conflict with old E2/E3 entries (17.3/20 and 0.575). The old E5 narrative claims 120 annotated contracts, not 120 human participants; no corresponding annotation/adjudication record has been established here. Neither values nor captions can be cited as current comparative or human precision. They are excluded from the main argument and require reconciliation before submission; preservation does not validate these claims.

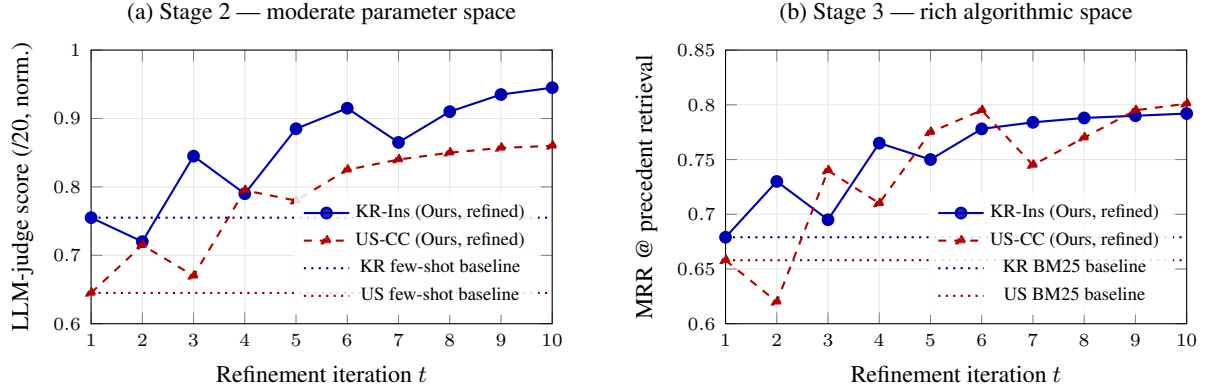


Figure 2: **Inherited Stage 2/3 iteration trace; not reproduced.** The original plot is retained for reconciliation, not as evidence on the fixed 3,000-source cohort. Its KR endpoints (18.9/20 and MRR 0.792) conflict with the inherited E2/E3 table values (17.3/20 and 0.575). No ten-iteration convergence or agreement with Tables 7–8 is established by the current run.

Dom.	Method	Trig.	Desc.	Query	Cov.	Total
KR-Ins	Zero-shot (multimodal)	1.1	1.0	1.0	1.2	4.3/20
	OCR – text (raw)	3.8	3.7	3.4	3.9	14.8/20
	Few-shot (FS)	3.9	3.6	3.8	3.8	15.1/20
	Ours (w/o refine)	3.9	3.8	3.7	4.0	15.4/20
	Ours	4.4	4.3	4.2	4.4	17.3/20
US-CC	Zero-shot (multimodal)	1.0	1.0	1.0	1.1	4.1/20
	OCR – text (raw)	2.9	2.9	2.8	3.2	11.8/20
	Few-shot (FS)	3.1	3.3	3.0	3.5	12.9/20
	Ours (w/o refine)	3.2	3.1	3.2	3.6	13.1/20
	Ours	4.2	4.3	4.1	4.6	17.2/20
US-Ln	OCR – text (raw)	2.7	3.1	1.0	1.7	8.6/20
	Few-shot (FS)	1.4	2.4	1.0	1.3	6.1/20
	Ours (w/o refine)	2.3	2.4	1.0	1.9	7.6/20
	Ours	2.1	3.3	3.3	1.7	10.4/20

Table 7: **E2 — Inherited input-format comparison; not reproduced.** These are earlier manuscript values, not measurements on the fixed 3,000 sources. The earlier draft attributes these four-axis scores to Claude Sonnet 4.5 and GPT-4o judges; that execution and averaging protocol are not verified here.

Dom.	Method	MRR	P@1	P@3	R@3	R@5
KR-Ins	BM25	0.492	0.286	0.254	0.714	0.810
	TF-IDF	0.431	0.190	0.222	0.667	0.762
	Emb.	0.447	0.238	0.238	0.690	0.786
	Hybrid	0.511	0.333	0.270	0.738	0.833
	Ours	0.575	0.381	0.349	0.810	0.905
US-CC	BM25	0.658	0.287	0.271	0.416	0.455
	TF-IDF	0.724	0.259	0.311	0.361	0.446
	Emb.	0.741	0.271	0.324	0.378	0.463
	Hybrid	0.769	0.346	0.314	0.434	0.511
	Ours	0.801	0.294	0.359	0.474	0.587
US-Ln	BM25	0.750	0.643	0.738	0.857	0.881
	TF-IDF	0.839	0.786	0.810	0.857	0.929
	Emb.	0.831	0.762	0.795	0.833	0.905
	Hybrid	0.836	0.786	0.786	0.857	0.929
	Ours	0.964	0.929	0.905	1.000	1.000

Table 8: **E3 — Inherited retrieval comparison; not reproduced.** These values do not describe the new 3,000-source experiment. KR-Ins: 17-precedent pool from data/precedents.json; US-CC: CourtListener API; US-Ln: in-domain retrieval pool of 663 covenant-tagged paragraph chunks from 216 SEC loan agreements. All $\uparrow$. The earlier draft labels the final row as a selected configuration; that selection and its relevance labels are not verified here.

Method	KR-Ins		US-CC		US-Ln	
	MRR	Δ	MRR	Δ	MRR	Δ
Single-agent	0.531	—	0.763	—	0.871	—
w/o Meta-review	0.548	+1.7	0.779	+1.6	0.903	+3.2
w/o Evolution	0.554	+2.3	0.784	+2.1	0.918	+4.7
Full (Ours)	0.575	+4.4	0.801	+3.8	0.964	+9.3

Table 9: **E4 — Inherited Stage 3 ablation; not reproduced.** The following interpretation is an unverified earlier claim: Removing either supporting agent degrades final MRR across all three tasks; the full three-agent loop achieves the largest gains. Δ denotes absolute MRR change relative to the single-agent baseline.

Method	Proposed	Precision $\uparrow$
Few-shot (FS)	4.2	0.531
Ours (w/o refine)	5.1	0.614
Ours	4.7	0.823

Table 10: **E5 — Inherited human assessment; trace not established.** Not verified precision or a current 3,000-source result. The earlier draft claims 120 sampled contracts and defines precision as validated candidates / total proposed; the annotation trace is not established.

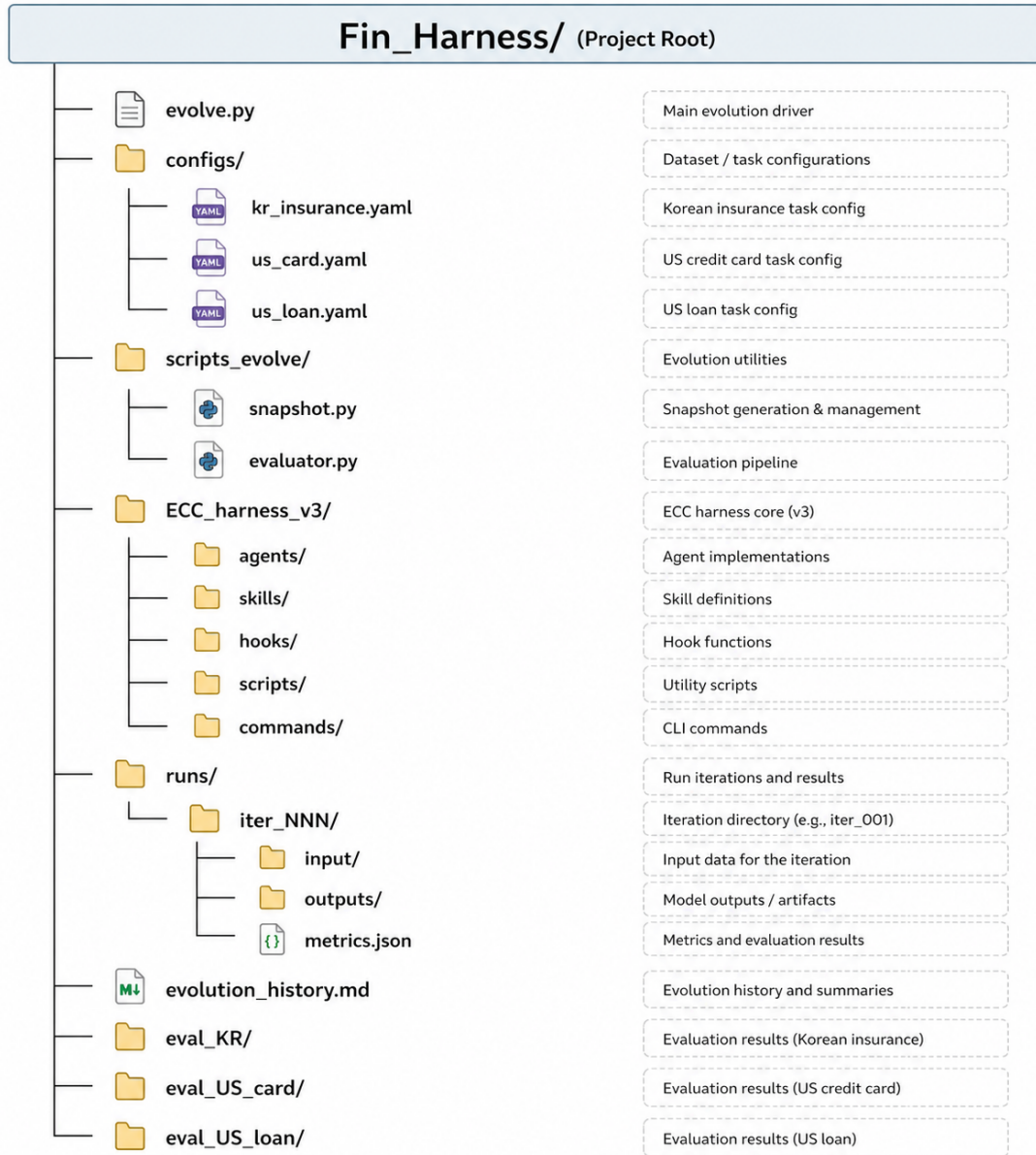


Figure 3: **Preserved historical directory diagram.** This image is a design artifact, not the executed file layout. The actual TXT harness and current Stage 1 entry point are described in Appendix A. No Stage 2/3 evolution trace is established.